

```
# Used to securely store your API key
from google.colab import userdata

# Retrieve the Groq API key from Colab secrets
GROQ_API_KEY = "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"

# Ensure the .env file exists and write the key to it
with open('.env', 'a') as f:
    f.write(f'GROQ_API_KEY="{GROQ_API_KEY}"\n')

print("GROQ_API_KEY written to .env file.")
```

GROQ_API_KEY written to .env file.

```
!pip install langgraph langchain_core langchain_groq typing
```

```
Requirement already satisfied: langgraph in /usr/local/lib/python3.12/dist-packages (1.2.1)
Requirement already satisfied: langchain_core in /usr/local/lib/python3.12/dist-packages (1.4.0)
Collecting langchain_groq
  Downloading langchain_groq-1.1.2-py3-none-any.whl.metadata (2.4 kB)
Collecting typing
  Downloading typing-3.7.4.3.tar.gz (78 kB)
 78.6/78.6 kB 6.1 MB/s eta 0:00:00

Preparing metadata (setup.py) ... done
Requirement already satisfied: langgraph-checkpoint<5.0.0,>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from langgraph) (4.1.0)
Requirement already satisfied: langgraph-prebuilt<1.2.0,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from langgraph) (1.2.0)
Requirement already satisfied: langgraph-sdk<0.4.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from langgraph) (0.3.0)
Requirement already satisfied: pydantic>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langgraph) (2.12.3)
Requirement already satisfied: xxhash>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from langgraph) (3.7.0)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from langchain_core) (1.33.0)
Requirement already satisfied: langchain-protocol>=0.0.14 in /usr/local/lib/python3.12/dist-packages (from langchain_core) (0.0.14)
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in /usr/local/lib/python3.12/dist-packages (from langchain_core) (0.3.45)
Requirement already satisfied: packaging>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from langchain_core) (26.2)
Requirement already satisfied: pyyaml<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langchain_core) (6.0.3)
Requirement already satisfied: tenacity!<8.4.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langchain_core) (8.4.0)
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from langchain_core) (4.7.0)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from langchain_core) (0.12.0)
Collecting groq<1.0.0,>=0.30.0 (from langchain_groq)
  Downloading groq-0.37.1-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: anyio<5,>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from groq<1.0.0,>=0.30.0->langchain_groq) (4.3.0)
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from groq<1.0.0,>=0.30.0->langchain_groq) (1.7.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from groq<1.0.0,>=0.30.0->langchain_groq) (0.23.0)
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (from groq<1.0.0,>=0.30.0->langchain_groq) (1.3.0)
Requirement already satisfied: jsonpointer>=1.9 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0->langchain_core) (2.3.0)
Requirement already satisfied: ormsgpack>=1.12.0 in /usr/local/lib/python3.12/dist-packages (from langgraph-checkpoint<5.0.0,>=4.1.0->langgraph) (1.12.0)
Requirement already satisfied: orjson>=3.11.5 in /usr/local/lib/python3.12/dist-packages (from langgraph-sdk<0.4.0,>=0.3.0->langchain_core) (3.11.5)
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langchain_core) (1.0.0)
Requirement already satisfied: requests>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langchain_core) (2.32.0)
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langchain_core) (0.23.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2.7.4->langchain_core) (0.6.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2.7.4->langchain_core) (2.41.4)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic>=2.7.4->langchain_core) (0.4.2)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5,>=3.5.0->groq<1.0.0,>=0.30.0->langchain_groq) (3.10.1)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->groq<1.0.0,>=0.30.0->langchain_groq) (2024.8.30)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->groq<1.0.0,>=0.30.0->langchain_groq) (1.0.6)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->groq<1.0.0,>=0.30.0->langchain_groq) (0.14.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0.0->langsmith) (3.4.0)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0.0->langsmith) (2.2.3)
Downloading langchain_groq-1.1.2-py3-none-any.whl (19 kB)
Downloading groq-0.37.1-py3-none-any.whl (137 kB)
137.5/137.5 kB 12.7 MB/s eta 0:00:00

Building wheels for collected packages: typing
  Building wheel for typing (setup.py) ... done
  Created wheel for typing: filename=typing-3.7.4.3-py3-none-any.whl size=26304 sha256=ada106945b2e45ddc68d3486922c175fca44
  Stored in directory: /root/.cache/pip/wheels/12/98/52/2bffe242a9a487f00886e43b8ed8dac46456702e11a0d6abef
Successfully built typing
Installing collected packages: typing, groq, langchain_groq
Successfully installed groq-0.37.1 langchain_groq-1.1.2 typing-3.7.4.3
```

```
import os, json
from dotenv import load_dotenv
from typing import TypedDict, List
from langgraph.graph import StateGraph, END
from langchain_core.messages import SystemMessage, HumanMessage
from langchain_groq import ChatGroq

load_dotenv()
```

```

llm = ChatGroq(
    temperature=0,
    model_name="llama-3.1-8b-instant",
    groq_api_key=GROQ_API_KEY
)

# --- shared state schema ---
class AgentState(TypedDict):
    goal: str
    tasks: List[str]
    results: List[str]
    critique: str
    approved: bool
    iterations: int

def planner(state: AgentState) -> AgentState:
    system = """You are a planning agent. Break the user's goal into
at most 5 concrete, actionable tasks. Respond ONLY with a
valid JSON array of strings. No preamble, no markdown."""

    messages = [
        SystemMessage(content=system),
        HumanMessage(content=f"Goal: {state['goal']}")
    ]
    response = llm.invoke(messages).content.strip()

    try:
        clean = response.replace("`json", "").replace("`", "").strip()
        tasks = json.loads(clean)
    except json.JSONDecodeError:
        tasks = [response] # fallback: treat whole response as one task

    print(f"\n[Planner] Generated {len(tasks)} tasks:")
    for i, t in enumerate(tasks): print(f" {i+1}. {t}")

    return {**state, "tasks": tasks}

initial_state: AgentState = {
    "goal": "Research and summarise the top 3 trends in agriculture for 2025",
    "tasks": [],
    "results": [],
    "critique": "",
    "approved": False,
    "iterations": 0
}
planner(initial_state)

```

```

[Planner] Generated 5 tasks:
1. Identify credible sources of agricultural research and trends
2. Research and gather information on the top 3 trends in agriculture for 2025
3. Analyze and categorize the gathered information into relevant trends
4. Evaluate and prioritize the top 3 trends based on relevance and impact
5. Create a concise summary of the top 3 trends in agriculture for 2025
{'goal': 'Research and summarise the top 3 trends in agriculture for 2025',
'tasks': ['Identify credible sources of agricultural research and trends',
'Research and gather information on the top 3 trends in agriculture for 2025',
'Analyze and categorize the gathered information into relevant trends',
'Evaluate and prioritize the top 3 trends based on relevance and impact',
'Create a concise summary of the top 3 trends in agriculture for 2025'],
'results': [],
'critique': '',
'approved': False,
'iterations': 0}

```

